

SIWA, Egypt

WMO: 624170

Lat: 29.200N

Lon: 25.483E

Elev: -49

StdP: 14.72

Time Zone: 2.00 (E02)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.9	43.0	23.4	17.6	70.5	27.1	21.0	66.1	21.0	61.9	18.9	61.4	1.2	200	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	106.3	70.2	103.7	69.8	101.4	69.7	74.1	93.9	73.3	93.2	72.4	92.5	6.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	106.4	81.6	67.6	101.2	81.6	66.3	96.6	81.8	37.5	93.6	36.7	93.0	36.0	92.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.7	(b) 14.8	(c) 12.6	(e) 36.7	(f) 111.3	(g) 2.1	(h) 2.2	(i) 35.2	(j) 112.9	(k) 34.0	(l) 114.2	(m) 32.8	(n) 115.5	(o) 31.2	(p) 117.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	56.0	58.9	65.9	73.0	80.2	86.2	88.6	88.7	84.6	76.1	65.8	58.7		
	DBStd	12.55	3.68	4.86	5.85	5.82	5.56	4.28	3.11	2.71	3.60	4.68	4.58	3.63		
	HDD50	2	2	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	758	279	183	56	3	0	0	0	0	0	0	39	198		
	CDD50	8632	189	249	494	690	935	1086	1196	1201	1039	809	474	270		
	CDD65	3913	1	11	85	244	470	636	731	736	589	344	63	3		
	CDH74	51561	33	152	939	2597	5819	8976	10796	10837	7587	3250	544	31		
	CDH80	29449	5	45	359	1169	3207	5503	6767	6690	4239	1342	122	1		
(14) Wind	WSAvg	4.5	4.5	5.6	5.9	6.0	5.3	4.7	4.4	4.2	4.0	3.2	2.9	3.8		
(15) Precipitation	PrecAvg	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecMax	0.9	0.3	0.3	0.3	0.6	0.2	0.0	0.0	0.1	0.1	0.3	0.6	0.3		
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.2	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.1	85.3	95.9	101.0	107.8	110.9	109.7	107.4	105.0	97.8	89.1	77.9		
		MCWB	55.9	57.1	61.2	64.1	67.6	70.8	71.2	72.1	70.0	68.7	64.9	57.4		
	2%	DB	72.0	78.2	88.6	95.3	101.9	105.6	105.6	104.1	101.6	93.3	83.2	73.2		
		MCWB	54.6	55.6	60.0	62.2	66.7	69.2	70.8	71.3	70.2	67.8	63.3	58.3		
	5%	DB	68.5	73.6	83.2	91.3	97.9	102.1	102.6	101.7	98.5	89.9	79.4	70.8		
		MCWB	54.2	55.1	58.3	62.2	65.8	68.7	70.2	71.0	69.9	67.4	62.0	57.3		
	10%	DB	66.3	69.8	78.5	86.9	94.2	99.0	100.1	99.5	95.8	87.3	76.7	68.7		
		MCWB	53.2	54.2	57.2	60.8	65.1	68.4	69.9	70.7	69.4	66.7	60.9	56.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	59.6	64.6	66.9	70.6	73.3	74.9	75.6	74.7	72.1	67.5	62.4		
		MCDB	71.0	77.5	84.0	93.8	98.3	99.2	95.6	95.3	94.3	88.9	82.6	72.2		
	2%	WB	56.6	57.6	61.7	64.6	68.7	71.8	73.6	74.4	73.2	70.2	65.3	60.2		
		MCDB	68.2	72.8	83.2	90.1	95.5	97.8	95.7	93.4	91.6	86.4	80.1	70.4		
	5%	WB	55.2	56.2	59.6	63.0	67.4	70.7	72.6	73.5	72.1	69.0	63.4	58.4		
		MCDB	66.7	70.4	79.7	86.5	93.1	96.0	94.4	92.2	90.3	85.3	77.0	69.0		
	10%	WB	53.9	54.9	58.1	61.6	66.2	69.7	71.7	72.7	71.1	67.9	61.9	56.9		
		MCDB	65.1	68.2	76.6	83.8	90.6	94.1	92.9	91.7	89.4	84.2	74.7	67.4		
(35) Mean Daily Temperature Range	MDBR		21.4	21.5	23.1	24.3	25.2	25.3	24.5	22.9	22.5	21.3	22.0	20.9		
	5% DB	MCDBR	24.7	27.5	29.4	30.6	29.7	28.6	27.4	25.5	25.8	24.1	25.0	23.1		
		MCWBR	12.8	11.6	10.6	10.2	9.3	7.9	7.3	7.1	7.7	8.8	11.4	12.3		
	5% WB	MCDBR	22.3	22.6	26.7	27.0	26.4	26.3	25.0	22.6	22.3	21.3	22.8	21.4		
		MCWBR	12.6	10.8	10.6	9.8	9.0	7.9	7.3	7.1	7.6	8.7	11.3	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.395	0.450	0.499	0.568	0.573	0.504	0.467	0.470	0.493	0.454	0.404	0.393		
	taud		2.021	1.856	1.732	1.616	1.631	1.800	1.909	1.909	1.845	1.943	2.039	2.039		
	Ebn at Noon		259	253	250	238	237	253	262	260	249	249	253	252		
	Edh at Noon		46	59	72	83	82	69	62	61	63	54	45	43		
(44) All-Sky Solar Radiation	RadAvg		1217	1492	1880	2178	2377	2542	2501	2318	2038	1668	1316	1130		
	RadStd		33	77	55	49	74	49	24	25	28	33	31	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page